

Dysregulation of Glycemic Control

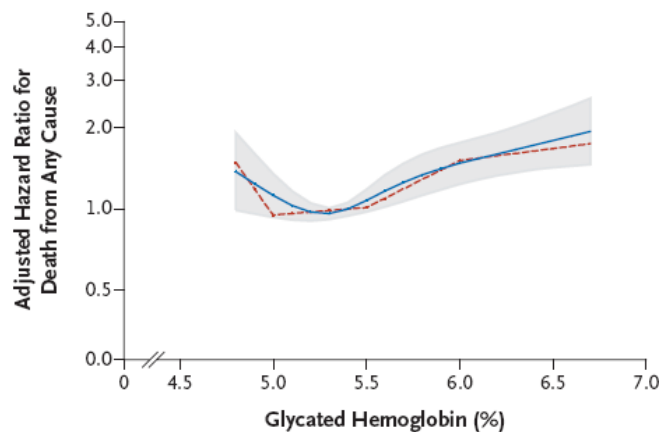
Instructor: Charles P. Scott, Ph.D. (charles.scott@jefferson.edu)
Block: Endocrine/Urology/Reproduction
Thread: Biochemistry
Conflict of Interest: Dr. Scott has no conflicts of interest to disclose.

Learning Objectives

1. Discuss the effect of type 1 diabetes on metabolism of glucose and fat, outline expected abnormalities in blood and urine laboratory values and suggest appropriate treatment.
2. Describe alterations to glucose counter-regulation in patients who have long-standing type 1 diabetes and explain why these alterations make patients more prone to hypoglycemia.
3. Summarize the abnormalities in endogenous glucose production, glycogen metabolism, and lipid metabolism that are often observed in patients who have type 2 diabetes.
4. Discuss the mechanisms and weaknesses of metformin, sulfonylurea/glinide drugs and GLP agonists as treatments for T2D.
5. Differentiate causes, diagnosis and treatment of post-prandial, fasting, ketotic and non-ketotic, hypoglycemia.

Practice questions

1. Investigators in internal medicine at Yale studied glycogen synthesis in non-diabetic controls and in volunteers who have MODY type 2, which is due to a mutation in the glucokinase gene. The mutant glucokinase has an elevated $S_{0.5}$ for glucose. Over the course of three meals, the MODY volunteers increased liver glycogen by only about 90% of the controls. How does MODY-2 contribute to this mild deficiency?
2. The following data refer to blood samples collected in 1990-1992. HbA_{1c} was determined around the year 2009. The presence of diabetes was also determined in 2009 by phone survey. The following graph was obtained:

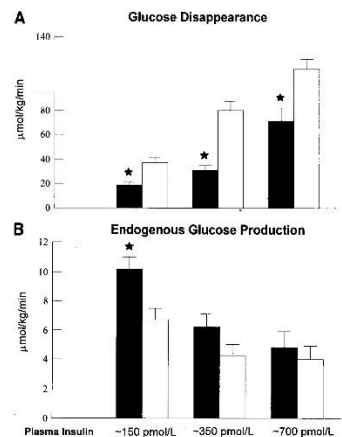


E Selvin et al., NEJM
362: 800-811, 2010.

An increase in HbA_{1c} predicted diabetes, and diabetes is associated with increased mortality. Give a reason why mortality was unexpectedly higher at an HbA_{1c} of 4.8% than 5.3%.

3. You are caring for a patient with type 1 diabetes. The patient has gained 10 pounds over the last year, but the “hemoglobin A_{1c}” is at a reasonably low 7.5%.” How and why did the patient gain weight?
4. The following questions are based on R. Basu et al. in Diabetes 53: 2042-2050, 2004. Patients with diabetes and control patients were studied. BMI = 32 ± 1 kg/m² in both groups. The volunteers fasted overnight. Patients with diabetes also received insulin to normalize blood glucose. Radioactively labeled glucose was infused overnight (to study glucose fluxes in the morning). As needed, glucose was infused to maintain blood glucose at 162 mg/dl (9 mM).

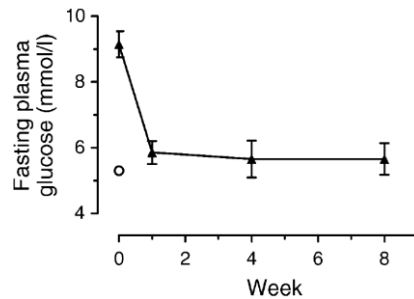
Insulin was infused to reach the approximate concentrations given in the figure below. The results were as follows:



- a. Regarding part A of the above figure: Where did glucose disappear to?
- b. Why did an increase in the concentration of insulin enhance glucose disappearance?
- c. Do the open or the closed bars represent the control patients?

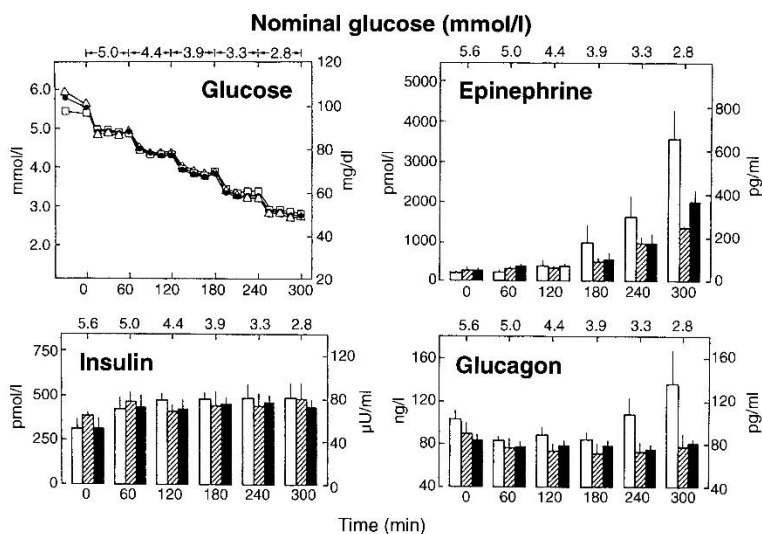
5. The following case is based on a case in DiabetesinControl.com. To solve the questions related to this case, you will need some knowledge of the diagnosis of diabetes and the pharmacology of drugs used to treat diabetes (these lessons will be taught today and on Thursday).
- a. A 20-year-old patient with obesity presents with symptoms that are compatible with diabetes. How would you determine whether this patient has diabetes?
 - b. This patient indeed had diabetes. What would be the most common way of initiating treatment for this patient?
 - c. Five years after diagnosis, the patient has a BMI of 45 kg/m^2 . They take metformin, a sulfonylurea, and a DPP-4 inhibitor. About once every two weeks, they have an episode of hypoglycemia. Which drug should you replace or prescribe in a smaller dose to avoid recurring hypoglycemia?
 - d. Four months later, the patient's HbA_{1C} level is 7.8% and the fasting C-peptide level is 6 ng/mL (normal range 0.78-1.89 ng/dL). How do you explain the elevated fasting C-peptide level?

6. Eleven patients diagnosed with type 2 diabetes were studied before a diet and at three points during an 8-week very-low-calorie diet (600 kcal/day; EL Lim *et al.*, *Diabetologia* 54: 2506-2514 (2011)). The following results were obtained:



- a. The concentration of free fatty acids increased from 0.69 ± 0.06 at baseline to 0.93 ± 0.05 mM at the end of week 1. Why? (FYI, in weight-matched control patients without diabetes, at baseline, the concentration of free fatty acids was 0.57 ± 0.07 mM.)
- b. After one week on the diet, the fasting plasma glucose concentration of the patients with diabetes was in the normal range, and patients behaved clinically like patients without diabetes. By contrast, the HbA_{1c} was $7.1 \pm 0.3\%$ (control patients without diabetes at baseline: $5.7 \pm 0.1\%$). Why was the HbA_{1c} of the patients with diabetes not close to that of the controls?

7. Using the **hyperinsulinemic stepped hypoglycemic clamp** technique, the authors assessed the response of patients with type 1 diabetes to hypoglycemia (PE Cryer, *Diabetologia* 45: 937-948, 2002).



- Why did the concentration of glucose drop during this experiment?
- In the above figure, the concentration of insulin in peripheral blood is comparable in patients with and without diabetes, because the investigators infused insulin such that the 3 groups would show similar concentrations.

For our purposes, we assume that the crosshatched and the filled bars represent the same patient population. Which volunteers have type 1 diabetes, those represented by the open columns, or those represented by the hatched/filled columns?

- Suppose that you stop the hyperinsulinemic hypoglycemic clamp at the 5-hour time-point. Which patient population recovers more quickly from the hypoglycemia? Why?

8. The antidiabetic drug metformin is widely prescribed to patients who have type 2 diabetes. Describe the alteration of the pathway that appears to be responsible for the major glucose-lowering effect of this drug.
9. Patients who have a hereditary disorder that impairs gluconeogenesis (e.g., a deficiency of glucose 6-phosphatase) develop lactic acidemia during a prolonged fast. Explain the pathogenesis of the lactic acidemia.
10. List the pathways that are more or less active as the plasma glucose concentration of a perfectly healthy person slips below 100 mg/dl after a meal. Then, explain which of these changes are blocked in a person who has insulin-induced hypoglycemia.

Overview

1. A near-complete lack of insulin leads to **diabetic ketoacidosis**, a condition of high concentrations of H^+ , ketone bodies, and glucose in the blood. The **low pH** is neurotoxic.
2. **Obesity** is associated with a diminished response to insulin and an increased risk of diabetes.
3. People who have **type 2 diabetes** have **excessive endogenous glucose production** due to an excessive rate of gluconeogenesis. About half of these people also have **dyslipidemia** and **hypertension**.
4. Older people under stress who have had type 2 diabetes for years are most at risk of developing the **hyperosmolar hyperglycemic state**, which is associated with extreme **dehydration** due to loss of glucose and water in the urine.

Terminology

The **major categories of diabetes** are:

Type 1 diabetes, which results from **autoimmune destruction** of pancreatic β -cells

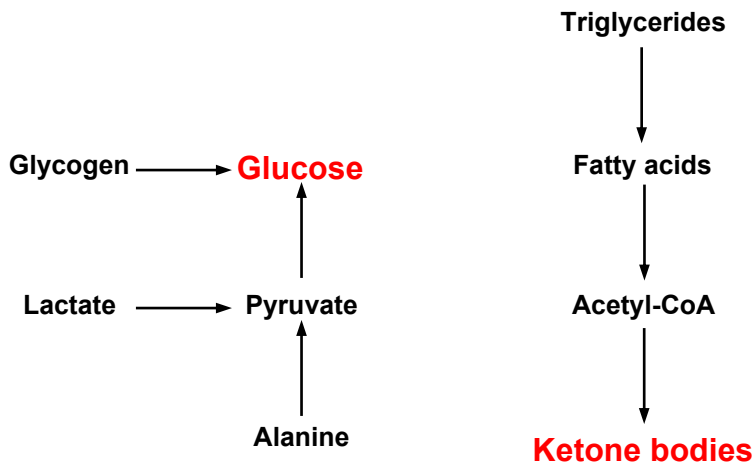
Type 2 diabetes, which encompasses a collection of diseases; it is most often characterized by a **reduced response to insulin** and a **reduced secretion of insulin** from pancreatic β -cells.

Gestational diabetes is diabetes that is first diagnosed during pregnancy.

(You will hear much more about all of these in other lectures this block).

A. Metabolism in Type 1 Diabetes

1. A **basal**, low concentration of **insulin** is needed at all times to **inhibit lipolysis** and thus prevent run-away ketone body production from excess acetyl-CoA (acetyl-CoA stems from excessive fatty acid β -oxidation).
(You will learn in your PHRM lectures that people with type 1 diabetes obtain a baseline concentration of insulin either
 - by injecting a bolus of a '**basal**', **long-acting insulin**, or
 - by using a wearable **pump** at a '**basal**' **rate** to move short-acting insulin into the subcutaneous space.)
2. A severe **deficiency of insulin** leads to **diabetic ketoacidosis (DKA)**, which is characterized by severe **hyperglycemia**, a high concentration of **ketone bodies** (acetoacetate and β -hydroxybutyrate) in the blood, and **acidosis**.
 - a. DKA is seen almost exclusively in **patients with type 1 diabetes**. These patients typically have no functioning β -cells. DKA can result when these patients have not injected insulin appropriately.
 - b. Pure DKA is rarely seen in patients with type 2 diabetes because they secrete enough insulin to inhibit lipolysis (see section C below).
3. In DKA, the lack of insulin causes the concentration of **glucose** to be high, because
 - a. **Glucose use is minimal**:
 - i. **GLUT-4** is NOT stimulated
 - ii. **Glycolysis** is NOT stimulated
 - iii. **Glycogen synthesis** is NOT stimulated
 - iv. **Fatty acids** are abundant and replace glucose as a source of energy
 - b. **Glucose production is maximal**:
 - i. **Glycogenolysis** in the liver is active (NOT inhibited by insulin)
 - ii. **Gluconeogenesis** in the liver and in the kidneys is active (NOT inhibited by insulin).



Production of glucose and ketone bodies in DKA. Glucose use is relatively low, and ketone body production far exceeds consumption, thus causing acidosis.

4. In DKA, the concentration of **ketone bodies** is high, because
 - a. there is no insulin to inhibit **lipolysis**, so the concentration of free **fatty acids** in the blood is high; and
 - b. the liver β -oxidizes more fatty acids than needed for ATP production and converts the remaining **acetyl-CoA** to **ketone bodies**.

5. In DKA, the rate of **ketone body production by the liver is greater than the rate of ketone body oxidation by other tissues** (due to the abundance of glucose as an energy source). As a result, the blood incurs an ongoing **acid load**, which **decreases blood pH**.
 - a. **Bicarbonate** buffers protons to yield H_2CO_3 and then CO_2 , which is exhaled.
 - b. The rate of CO_2 exhalation far exceeds the rate of bicarbonate production by the kidneys, leading to a low concentration of **bicarbonate**.
 - i. To produce bicarbonate, the kidneys convert glutamine to bicarbonate and ammonium ions. The HCO_3^- is transported into the blood, and NH_4^+ ends up in the urine (think of this as H^+ or acid secretion).

6. In DKA, the **urine** contains **glucose** and **ketone bodies**. These osmolytes are extremely hygroscopic, so they cause diuresis, which leads to **severe dehydration**.

- a. Glucose spills into the urine when the blood contains more than about **180 mg/dl glucose**.
 - b. Ketone bodies spill into the urine when blood contains more than about **5 mM ketone bodies**.
7. In untreated DKA, it is the **acidosis** that **brings patients to medical attention**. A low pH is a major threat to the function of the CNS.

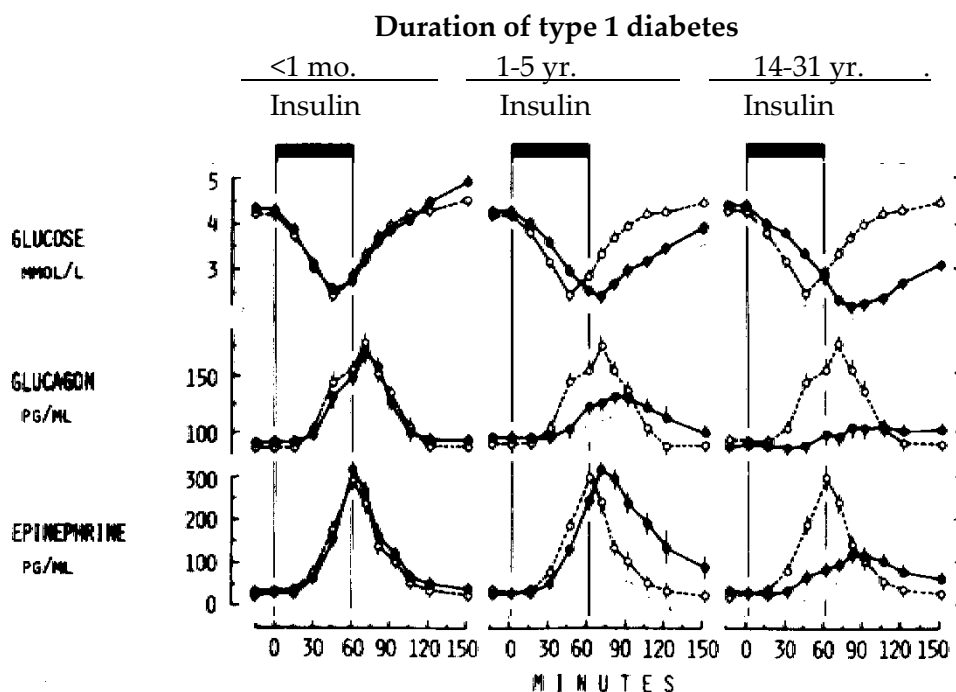


8. DKA is treated in part with **insulin**, which reduces the rate of lipolysis and thus reduces the rate of fatty acid β -oxidation and ketone body production in the liver. Decreasing acid production enables blood pH to normalize.
9. Keep in mind that in a person who has had type 1 diabetes for at least several months, DKA develops in a matter of **hours** to less than a day (it largely depends on the duration of action of administered insulin).
10. Typical **laboratory abnormalities in DKA** are as follows:
- | | |
|-----------------------------------|-------------------|
| Plasma glucose: | >250 mg/dl |
| Bicarbonate: | < 18 mEq/l |
| pH of arterial blood: | <7.30 |
| Ketone bodies in plasma or urine: | Markedly elevated |
| Anion gap: | Elevated |

B. Loss of glucose counter-regulation in T1D

People with type 1 diabetes develop an impairment of glucose counter-regulation.

- a. As the duration of type 1 diabetes increases, **glucagon secretion** in response to hypoglycemia is lost; later, **epinephrine secretion** in response to hypoglycemia is also lost:



Patients with type 1 diabetes of long duration release too little glucagon and epinephrine in response to insulin-induced hypoglycemia. Open circles with dashed lines (same data repeated in all 3 columns); nondiabetic volunteers. Filled circles and solid lines: three groups of type 1 diabetic patients (<1 month, 1-5 years, or 14-31 years since diagnosis). Insulin was infused intravenously from 0 to 60 min. Means and SEMs are shown.
G Bolli et al., Diabetes 32: 134-141, 1983.

- b. Some caveats:
 - i. Since these data were acquired, the management of diabetes has changed dramatically.
 - ii. While the above measurements have not been repeated, the consensus is that the findings still apply.
 - iii. Paradoxically, an episode of hypoglycemia diminishes the counter-regulatory response to a subsequent episode of hypoglycemia.
- c. Hypoglycemia is acutely dangerous and therefore a limiting factor in the use of insulin.

C. Metabolism in Type 2 Diabetes

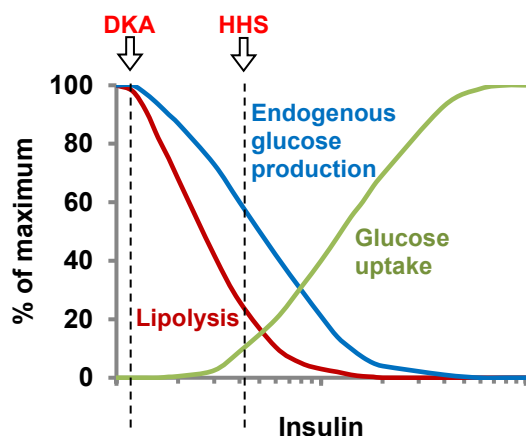
1. Most patients with type 2 diabetes are **insulin-resistant**, and their β -cells secrete **insufficient insulin to control blood glucose**. (Thereby, they may secrete less or more insulin than a person who is insulin-sensitive and not diabetic.)
2. In type 2 diabetes, the inadequate concentration of **insulin** leads to the following:
 - a. Reduced **glucose use** due to
 - i. Reduced glycogen synthesis in the fed state
 - b. Excessive endogenous **glucose production** due to
 - i. Inadequate suppression of **glycogenolysis**
 - ii. Inadequate suppression of **gluconeogenesis**
 - c. Inadequate activation of adipose lipoprotein lipase, leading to decreased removal of **triglycerides** from circulating lipoprotein particles
 - d. Inadequate suppression of **lipolysis**, which can lead to excessive production of **VLDL**.
3. The term **metabolic syndrome** was coined to study the association between insulin resistance, glucose intolerance, dyslipidemia, and hypertension with an increased risk of cardiovascular disease. Despite major problems and a retraction by professional diabetes associations, the term has become a favorite of physicians and even a billable ICD code.
 - a. The most common definitions of metabolic syndrome involve meeting a subset of the following criteria:
 - Plasma glucose concentration typical of diabetes
 - Increased waist circumference or BMI
 - Low HDL
 - High triglycerides
 - Hypertension
 - b. In terms of biochemistry, the term metabolic syndrome is rather useless.
4. About half of patients who have type 2 diabetes have
High **triglycerides** (>200 mg/dl) and
Low **HDL**-cholesterol (<40 mg/dl).
LDL-C is often normal, yet **LDL are small and dense** so that they are more readily **oxidized** and more readily end up in **atherosclerotic plaques**.
5. Prevailing explanations for the **dyslipidemia** are as follows:
 - a. Due to an increased size of the adipose tissue, the net rate of **lipolysis** increases, and more fatty acids end up in the liver.

- b. De novo **fatty acid synthesis** in the liver is increased several fold (in part because a high concentration of glucose stimulates it independent of insulin).
 - c. **VLDL production** in the liver is increased.
 - d. Almost all people with obesity have a fatty liver, and fatty liver is associated with **liver insulin resistance**.
 - e. Insulin fails to inhibit the conversion of smaller VLDL2 particles to larger **VLDL1** particles. Larger lipid droplets in hepatocytes also favor the conversion of VLDL2 to VLDL1. Hence, there is increased production of the large VLDL1 particles (large VLDL are more atherogenic).
 - f. Insulin fails to adequately stimulate **lipoprotein lipase** activity in adipose tissue.
 - g. CETP exchanges more triglycerides for cholesterol esters in HDL and LDL, thus adding triglycerides to HDL and LDL. After hepatic lipase removes the triglycerides, **HDL** are small and lost, while **LDL** are small, dense, and more atherogenic.
6. FYI, much less is known about the pathogenesis of **dyslipidemia** in patients who have **type 1 diabetes**. A patient with type 1 diabetes and obesity has similar abnormalities of metabolism as a patient with type 2 diabetes and obesity. However, most patients with type 1 diabetes are not obese and therefore tend to have lesser abnormalities of lipid metabolism. On the other hand, patients with type 1 diabetes often have more and longer periods of poor control of plasma glucose than do patients with type 2 diabetes.
6. In patients with type 1 diabetes, various abnormalities of **glycogen** metabolism and **gluconeogenesis** have been described, and these seem to be linked to the quality of blood glucose control these patients achieve. Often, patients with type 1 diabetes seem to have similar abnormalities to patients with type 2 diabetes in that they synthesize and break down a reduced amount of glycogen, and they tend to have excessive gluconeogenesis that is not suppressed by insulin.
8. **Most** forms of type 2 diabetes show **complex genetics**. More than 100 predisposing traits are known, but each trait only has a small effect, and the presence of traits cannot yet be used to predict diabetes.
9. At most a few % of all patients with diabetes have a **monogenic** form of diabetes that almost always leads to impaired secretion of insulin. These forms are often classified as follows:
- Persistent neonatal diabetes** (onset within 1st year of life)
 - Transient neonatal diabetes** (onset within 1st year of life)
 - Maturity-onset diabetes of the young (MODY)**
- Genetically, there is overlap between these classes. Genetic testing guides the therapy.

- a. All persons who develop **diabetes before age 1 year** should be checked for mutation status. Worldwide resource (testing is currently free):
<https://www.diabetesgenes.org/about-neonatal-diabetes/genetic-testing-for-neonatal-diabetes/>
- b. There are several domestic services that test for monogenic diabetes, for instance <https://dnatesting.uchicago.edu/> and <https://www.questdiagnostics.com/home/physicians/testing-services/condition/diabetes/test-menu.html>

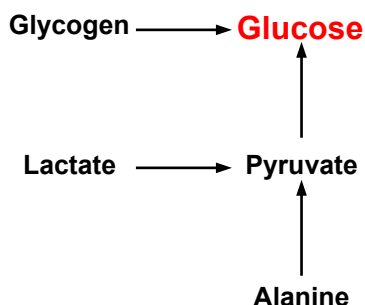
D. Metabolism in the Hyperosmolar Hyperglycemic State (HHS)

1. The hyperosmolar hyperglycemic state results from **days** of a pathogenic, low concentration of insulin. It is most commonly seen in elderly, socially isolated patients who have had **type 2 diabetes** for many years and who are experiencing a **stress** (including infection) that increases the concentrations of counter-regulatory hormones. Less commonly, HHS is seen in children and young adults as a first presentation of type 1 diabetes.
2. The **hyperosmolar hyperglycemic state (HHS)** is characterized by extreme **dehydration** due to extreme **hyperglycemia**. What distinguishes this from DKA is lack of pronounced ketoacidosis.
3. In pure HHS, there is enough insulin to suppress **lipolysis**, but not enough to suppress **endogenous glucose production**.



DKA vs. HHS: a question of the concentration of insulin.

(In part based on: M Stumvoll and S Jacob, Exp Clin Endocrinol Diabetes 107: 107-110, 1999.)



Production of glucose in HHS. Lipolysis and ketone body synthesis are suppressed by insulin.

- a. In HHS (as in DKA), **glucose use** is low due to the low concentration of insulin.
 - b. In HHS (as in DKA), **glucose production** is high due to the low concentration of insulin. However, in HHS (unlike DKA) the concentrations of **glucagon**, **epinephrine**, and **cortisol** are also markedly elevated, and all these hormones stimulate gluconeogenesis.
 - c. The high concentration of glucose in the blood leads to **glucosuria** (the threshold for the kidneys is about 180 mg/dl) and thus to severe **dehydration**.
 - d. In HHS (unlike DKA), the patient's β -cells secrete enough insulin to attenuate **lipolysis** and thus prevent excessive ketone body production.
 - e. HHS (unlike DKA in a person who is known to have type 1 diabetes) takes several **days** to develop.
4. In pure HHS, it is the severe **dehydration** that **brings patients to medical attention**.
 5. Severe dehydration causes mental status changes (confusion) or loss of consciousness because hypovolemia causes poor oxygen perfusion, particularly to the brain and vital (**hypovolemic hypoxia** or even hypovolemic shock). Patients displaying symptoms of severe dehydration (e.g., HHS) require emergent treatment with **fluids** to protect neural and function. **Insulin** may be given to reduce glucose production.



organs

organ

6. DKA and HHS are **extremes of a spectrum** of metabolic derangements that are seen in patients with diabetes. The prevailing concentrations of insulin determine where on the spectrum a patient lands.



- a. Pure DKA is most common among patients with type 1 diabetes.
 - b. Pure HHS is most common among patients who have type 2 diabetes.
7. Typical laboratory abnormalities in DKA and HHS are as follows:

Lab value	DKA	HHS
Plasma glucose, mg/dl	>250	>600
Osmolality, mOsm/kg	<320	>320
Bicarbonate, mEq/l	<15	>15
pH in arterial blood	<7.30	>7.30
Ketone bodies in serum or plasma	Markedly elevated	Low or absent

Derived from: Harrison's Principles of Internal Medicine, 19e (2015); Current Diagnosis and Treatment: Emergency Medicine, 8e (2017); AE Kitabchi et al., Diabetes Care (2009);32:1335-1343.

E. Approach to Hypoglycemia

1. Signs and symptoms of hypoglycemia: nervousness, anxiety, irritability, sweating, lightheadedness, dizziness, headache, impaired vision, confusion, seizures, loss of consciousness.
2. A suspicion of hypoglycemia can be confirmed with point-of-care fingerstick blood glucose.
3. If hypoglycemia needs to be replicated, it can be done with a mixed meal test for postprandial hypoglycemia and with up to a 3-day fast for fasting hypoglycemia.
 - a. Hypoglycemia is confirmed if symptoms are relieved when plasma glucose returns to normal

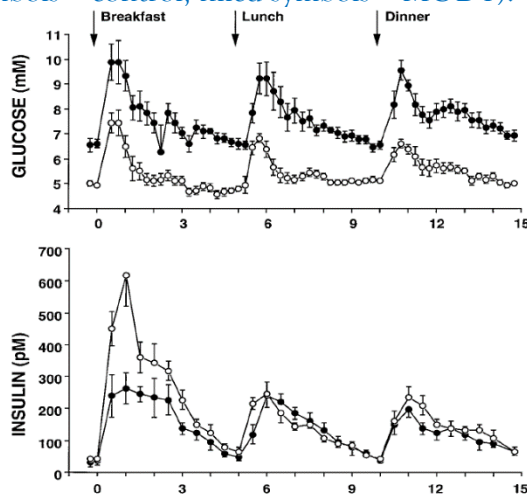
4. One approach to elucidating the cause of a patient's hypoglycemia – more often used by pediatricians – is to estimate the concentration of ketone bodies. This yields two categories of hypoglycemia:
 - a. **Non-ketotic hypoglycemia** (also called **hypoketotic hypoglycemia**). This can be caused by:
 - i. Excess insulin (insulin inhibits lipolysis, which in turn minimizes fatty acid oxidation and ketone body synthesis), which can be due to
 - Excess injected insulin
 - Excess of a sulfonylurea or glinide (drugs that close K_{ATP} and thus stimulate insulin secretion)
 - Insulinoma or congenital hypersecretion of insulin
 - ii. Deficient fatty acid oxidation, e.g., due to inherited MCAD deficiency
 - b. **Ketotic hypoglycemia**. This can be caused by:
 - i. Deficient gluconeogenesis (e.g., G6Pase deficiency)
 - ii. Deficient glucose production from glycogen, but this is rare and it is transient since gluconeogenesis makes up for the lack of glucose production from glycogen
5. Another approach is to determine when the patient develops hypoglycemia:
 - a. **Postprandially** (also called **reactive hypoglycemia**). Causes include congenital disorders (galactosemia, HFI), gastric bypass surgery.
 - b. In the **fasting state**. This may be caused by:
 - i. Insulinoma
 - ii. Deficient gluconeogenesis
 - iii. Deficient glycogenolysis (but gluconeogenesis can reduce this problem)
3. An **enzyme deficiency** in gluconeogenesis has the following consequences:
 - a. **Hypoglycemia** during a prolonged fast or intense exercise.
 - b. **Lactic acidosis** during a prolonged fast or intense exercise.
4. A deficiency of **glucose 6-phosphatase (von Gierke disease)** is the most common heritable disorder of **gluconeogenesis** (1:100,000); it is also a **glycogen storage disease (GSD type 1 = GSD1)**.
Glucose 6-phosphatase converts glucose 6-phosphate to glucose in the liver and in the kidneys.
 - a. G6Pase-deficient patients must avoid fasting. They typically drink **uncooked cornstarch** in water every few hours so that they have an ongoing source of glucose from the digestion of starch.

- a. Do not confuse glucose 6-phosphatase (**G6Pase**) deficiency and glucose 6-phosphate dehydrogenase (**G6PD**) deficiency (see Blood Block, Week 1, “RBC Metabolism & Anemia”).

Answers to practice questions

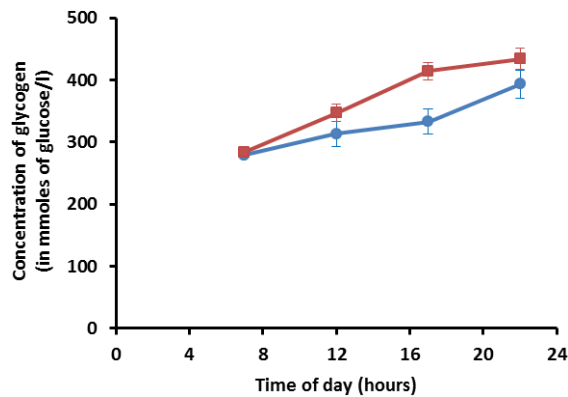
- Investigators in internal medicine at Yale studied glycogen synthesis in non-diabetic controls and in volunteers who have MODY type 2, which is due to a mutation in the glucokinase gene. The mutant glucokinase has an elevated $S_{0.5}$ for glucose. Over the course of three meals, the MODY volunteers increased liver glycogen by only about 90% of the controls. How does MODY-2 contribute to this mild deficiency?

Answer: Volunteers who have MODY-2 are heterozygous for a mutant **glucokinase** that has an increased $S_{0.5}$ for glucose. This increases the threshold for glucose-induced insulin secretion from pancreatic β -cells. Hence, these volunteers have an increased concentration of **glucose** and a decreased concentration of **insulin**, as is apparent from the following results (open symbols = control, filled symbols = MODY):

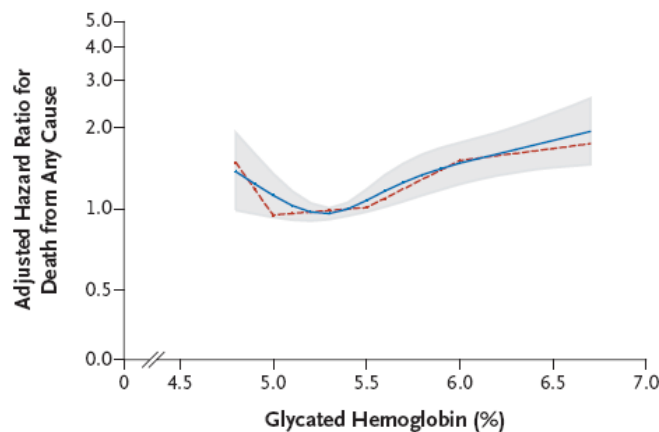


- In the MODY-2 volunteers, the liver, like the pancreatic β -cells, contain mutant **glucokinase** that has an elevated threshold for activation by glucose. Hence, this decreases the formation of glucose 6-phosphate and the rate of glycogen synthesis.
- The MODY-2 volunteers secrete somewhat less **insulin**, and this leads to reduced stimulation of glycogen synthase and hence decreased glycogen synthesis.
- Blood in the hepatic portal vein in the MODY-2 volunteers likely also has a higher concentration of **glucose**. If so, this would tend to increase glucokinase activity, the concentration of glucose 6-phosphate, and the rate of glycogen synthesis.

- d. It appears that the processes described in a. and b. together are more important than the process described in c (red curve below = control, blue curve = MODY).



2. The following data refer to blood samples collected in 1990-1992. HbA_{1c} was determined around the year 2009. The presence of diabetes was also determined in 2009 by phone survey. The following graph was obtained:



E Selvin et al., NEJM
362: 800-811, 2010.

An increase in HbA_{1c} predicted diabetes, and diabetes is associated with increased mortality. Give a reason why mortality was unexpectedly higher at an HbA_{1c} of 4.8% than 5.3%.

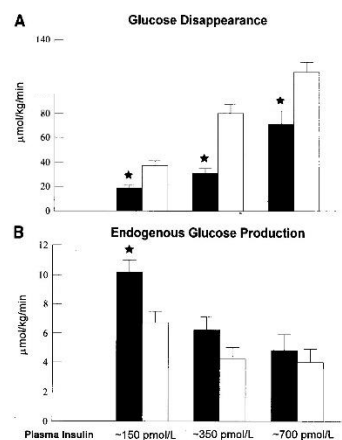
Answer: A significant portion of patients who had an HbA_{1c} below 5% most likely had increased turnover of red blood cells. Blood loss, sickle cell anemia, β -thalassemia, etc. can be associated with an increased risk of death.

3. You are caring for a patient with type 1 diabetes. The patient has gained 10 pounds over the last year, but the “hemoglobin A_{1c}” is at a reasonably low 7.5%. How and why did the patient gain weight?

Answer: The patient consumed too many calories yet injected the correct amount of insulin to control the concentration of blood glucose. It's the same scenario as for a patient without diabetes, except that the patient with diabetes had to control the supply of insulin themselves, since their pancreatic beta-cells were essentially absent.

4. The following questions are based on R. Basu et al. in Diabetes 53: 2042-2050, 2004. Patients with diabetes and control patients were studied. BMI = 32 ± 1 kg/m² in both groups. The volunteers fasted overnight. Patients with diabetes also received insulin to normalize blood glucose. Radioactively labeled glucose was infused overnight (to study glucose fluxes in the morning). As needed, glucose was infused to maintain blood glucose at 162 mg/dl (9 mM).

Insulin was infused to reach the approximate concentrations given in the figure below. The results were as follows:



- a. Regarding part A of the above figure: Where did glucose disappear to?

Glucose was sequestered by tissues that use glucose (RBC, brain, liver, heart, muscle, etc.). Almost none went into the urine, as the threshold for leakage from blood into the urine is about 180 mg glucose/dl.

- b. Why did an increase in the concentration of insulin enhance glucose disappearance?

Insulin stimulates glucose uptake into muscle and fat tissue, and it stimulates glucose use in the liver (by glycolysis, fatty acid synthesis, and glycogenesis).

- c. Do the open or the closed bars represent the control patients?

The open bars (insulin more readily stimulates glucose disappearance and more readily inhibits glucose production, i.e., the patients with the open bars are more insulin-sensitive than the patients with the filled bars).

5. The following case is based on a case in DiabetesinControl.com. To solve the questions related to this case, you will need some knowledge of the diagnosis of diabetes and the pharmacology of drugs used to treat diabetes (these lessons will be taught later today and on Thursday).

- a. A 20-year old patient with obesity presents with symptoms that are compatible with diabetes. How would you determine whether this patient has diabetes?

Measure fasting plasma glucose. If it is 126 mg/dl or more on two occasions, they have diabetes. Similarly, if their HbA_{1c} is $\geq 6.5\%$ in two measurements, they have diabetes.

- b. This patient indeed had diabetes. What would be the most common way of starting treatment for this patient?

Advise weight loss and exercise; prescribe metformin.

- c. Five years after diagnosis, the patient has a BMI of 45 kg/m². They take metformin, a sulfonylurea, and a DPP-4 inhibitor. About once every two weeks, they have an episode of hypoglycemia. Which drug should you replace or prescribe in a smaller dose to avoid recurring hypoglycemia?

The sulfonylurea drug, which stimulates insulin secretion even during hypoglycemia. By themselves, metformin and DPP-4 inhibitors do not cause hypoglycemia, but in the presence of a sulfonylurea, they may exacerbate hypoglycemia.

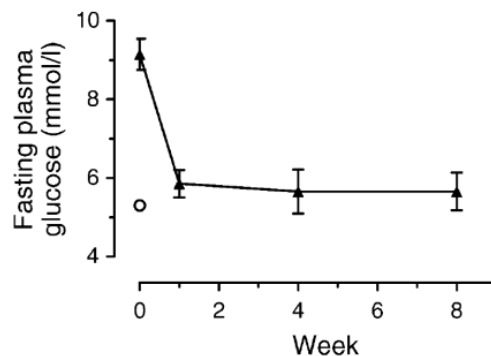
- d. Four months later, the patient's HbA_{1c} level is 7.8% and the fasting C-peptide level is 6 ng/mL (normal range 0.78-1.89 ng/dL). How do you explain the elevated fasting C-peptide level?

Due to the severe obesity (BMI of 45 kg/m²), the patient is very insulin-resistant. To compensate for the resistance, their pancreatic β -cells secrete more insulin and therefore more C-peptide. However, their β -cells have insufficient capacity to secrete insulin, hence the loss of control over the concentration of glucose in the blood.

Note that among people who have type 2 diabetes, some secrete more insulin than people who don't have diabetes, while others secrete less. Regardless of the amount of insulin that people with type 2 diabetes secrete, it is not enough to maintain a normal concentration of glucose in plasma.

6. Eleven patients diagnosed with type 2 diabetes were studied before a diet and at three points during an 8-week very-low-calorie diet (600 kcal/day). EL Lim *et al.*, *Diabetologia* 54: 2506-2514 (2011).

The following results were obtained:



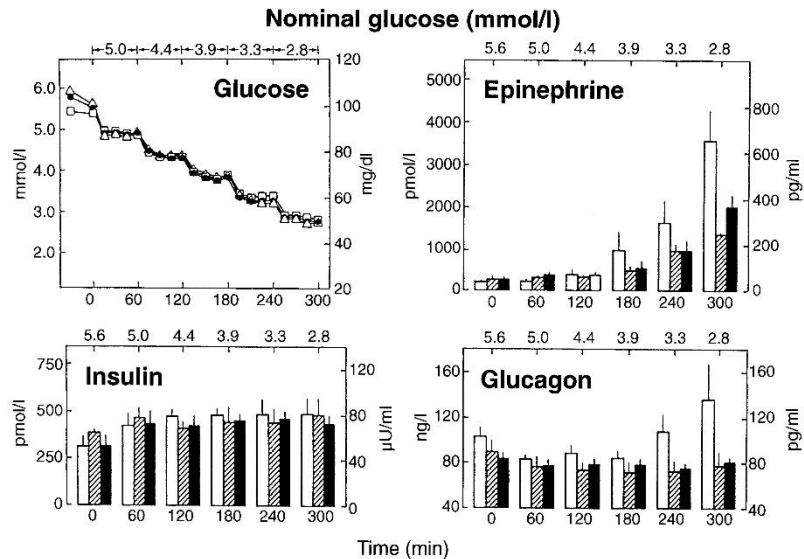
- a. The concentration of free fatty acids increased from 0.69 ± 0.06 at baseline to 0.93 ± 0.05 mM at the end of week 1. Why? (FYI, in weight-matched controls without diabetes, at baseline, the concentration of free fatty acids was 0.57 ± 0.07 mM.)

The patients were on a hypocaloric diet, and the insulin/glucagon plus epinephrine ratio was such that lipolysis and fatty acid β -oxidation were increased to make up for the deficit in calorie intake.

- b. After one week on the diet, the fasting plasma glucose concentration of the patients with diabetes was in the normal range, and patients behaved clinically like patients without diabetes. By contrast, the HbA_{1c} was $7.1 \pm 0.3\%$ (control patients without diabetes at baseline: $5.7 \pm 0.1\%$). Why was the HbA_{1c} of the patients with diabetes not close to that of the controls?

Red blood cells have a lifetime of about 120 days. Hence, hemoglobin also has a lifetime in the blood of about 120 days. After 1 week, most of the hemoglobin that had been exposed to an elevated concentration of glucose was still circulating in the blood. Due to the long lifetime of hemoglobin, it takes a long time for the HbA_{1c} to reflect a sudden change in mean blood glucose concentration.

7. Using the **hyperinsulinemic stepped hypoglycemic clamp** technique, the authors assessed the response of patients with type 1 diabetes to hypoglycemia (PE Cryer, *Diabetologia* 45: 937-948, 2002).
- 8.



- a. Why did the concentration of glucose drop during this experiment?

The patients were infused with insulin, which promotes removal of glucose from the blood.

- b. In the above figure, the concentration of insulin in peripheral blood is comparable in patients with and without diabetes because the investigators infused insulin such that the 3 groups would show similar concentrations.

For our purposes, we assume that the crosshatched and the filled bars represent the same patient population. Which volunteers have type 1 diabetes, those represented by the open columns, or those represented by the hatched/filled columns?

It must be those represented by the hatched/filled columns are long-standing patients with type-1 diabetes because these groups lack a glucagon response to hypoglycemia and show impaired epinephrine secretion in response to hypoglycemia. This is typical of patients who have had type 1 diabetes for several years.

- c. Suppose that you stop the hyperinsulinemic hypoglycemic clamp at the 5-hour time-point. Which patient population recovers more quickly from the hypoglycemia? Why?

The group represented by the open bars, because they secrete more epinephrine and glucagon.

8. The antidiabetic drug metformin is widely prescribed to patients who have type 2 diabetes. Describe the alteration of the pathway that appears to be responsible for the major glucose-lowering effect of this drug.

Metformin inhibits complex I of oxidative phosphorylation, which decreases ATP production and increases AMP concentration, thereby activating AMP-dependent protein kinase (AMPK). Both AMP and AMPK inhibit gluconeogenesis.

9. Patients who have a hereditary disorder that impairs gluconeogenesis (e.g., a deficiency of glucose 6-phosphatase) develop lactic acidemia during a prolonged fast. Explain the pathogenesis of the lactic acidemia.

Gluconeogenesis in the liver is a major consumer of lactic acid. Lactic acid is in equilibrium with pyruvate, thanks to the activity of lactate dehydrogenase. In the fasting state, pyruvate dehydrogenase, the enzyme that converts pyruvate to acetyl-CoA, is rather inactive, which directs pyruvate to glucose via gluconeogenesis. In G6P deficiency, this pathway is blocked. A block in G6Pase leads to a high concentration of G6P, which is in equilibrium with F6P. A high concentration of F6P activates phosphofructokinase 1, which generates F1,6BP. F1,6BP in turn activates pyruvate kinase. Hence, a deficiency of G6Pase leads to energy-wasting vicious cycles of glycolysis and gluconeogenesis with virtually no net consumption of pyruvate; hence, lactate accumulates.

10. List the pathways that are more or less active as the plasma glucose concentration of a perfectly healthy person slips below 100 mg/dl after a meal. Then, explain which of these changes are blocked in a person who has insulin-induced hypoglycemia.

In a perfectly healthy person whose plasma glucose concentration drops, insulin secretion decreases and glucagon secretion increases. The altered insulin/glucagon ratio leads to a decrease in glucose use, increased endogenous glucose production, increased lipolysis and hence increased fatty acid beta-oxidation. All of these processes help increase the plasma glucose concentration.

In a person who has insulin-induced hypoglycemia, the insulin/glucagon ratio changes in the opposite direction and thus leads to increased glucose use, decreased endogenous glucose production, decreased lipolysis and hence decreased fatty acid beta-oxidation. This person will likely have a relatively low plasma glucose concentration and will consequently also activate epinephrine secretion. Epinephrine then activates lipolysis and hence permits increased fatty acid beta-oxidation. This will lead to a lessening of the hypoglycemia.